



WEEK 6: ASSIGNMENT 6

Question 1:

A three-phase 11 kV / 66 kV delta/star connected power transformer is protected by a differential protection scheme. The CTs on the 11 kV (delta) side have a ratio of 1500/5 A. Determine the required CT ratio on the 66 kV (star) side.

- a) 250 : 5 A
- b) 250 : $5\sqrt{3}$ A
- c) 125 : $5\sqrt{3}$ A
- d) none of these

Answer: (b)

Solution:

Transformation ratio = $66/11 = 6$. Primary current on 66 kV side = $1500/6 = 250$ A. The 11 kV winding is delta so its CTs are star-connected; the 66 kV winding is star so its CTs are delta-connected and carry a secondary current of $5\sqrt{3}$ A. Hence the required CT ratio on the 66 kV side = 250: $5\sqrt{3}$ A

Question 2:

A percentage-biased differential relay protecting a generator has a pick-up current of 0.25 A, a bias slope of 20%, and a CT ratio of 800/1 A. During a through (external) fault, the currents in CT1 and CT2 are 720 A and 640 A, respectively. State whether the relay will operate or not.

- a) Relay will operate
- b) Relay is on the verge of operation
- c) Relay will not operate
- d) Data is insufficient

Answer: (c)

Solution:

$i_1 = 720/800 = 0.9$ A; $i_2 = 640/800 = 0.8$ A. Operating current = $|i_1 - i_2| = 0.1$ A. Restraining current = $(i_1 + i_2)/2 = 0.85$ A. Bias threshold = $0.20 \times 0.85 = 0.17$ A. Since the operating current (0.1 A) is below both the bias threshold (0.17 A) and the pick-up current (0.25 A), the relay will not operate.

Question 3:

A 13.8 kV star-connected synchronous generator is grounded through a neutral resistance of 4.5 Ω . The earth-fault relay is fed by a CT of ratio 6000/5 A and is set to pick up at 10% of its rated current. Determine the percentage of the stator winding, measured from the neutral end, that remains unprotected.

- a) 28.74%
- b) 23.45%
- c) 39.12%
- d) 33.89%

Answer: (d)

Solution:

Phase voltage = $13800/\sqrt{3} = 7967.4$ V.

Relay pick-up current referred to the CT primary = $0.5 \times (6000 / 5) = 0.10 \times 6000 = 600$ A.

The unprotected fraction x satisfies $x (V_{\text{phase}} / Z_n) = I_{\text{pickup}}$.

So, $x = (I_{\text{pickup}} \times Z_n) / V_{\text{phase}}$

$x = (600 \times 4.5) / 7967.4 = 0.3389$.

About 33.89% of the winding from the neutral end is unprotected.

Question 4:

In transformer differential protection, an interposing CT (or tap setting) is primarily used to

- a) increase the fault current.
- b) correct the ratio mismatch and phase-shift between the line CTs on the two sides.
- c) provide earth-fault protection only.
- d) replace the Buchholz relay.

Answer: (b)

Question 5:

The percentage (biased) differential relay is preferred over a plain (Merz-Price) differential relay because it

- a) remains stable against mal-operation due to CT ratio mismatch and CT saturation during through faults.
- b) operates faster for all faults.
- c) needs only one current transformer.
- d) does not require any setting.

Answer: (a)

Question 6:

Restricted earth-fault (REF) protection is applied to a transformer winding mainly to

- a) protect against inter-turn faults only.
- b) provide sensitive and high-speed protection for earth faults near the neutral end of the winding.
- c) detect magnetizing inrush.
- d) protect against overvoltage.

Answer: (b)

Question 7:

Transformer differential protection is restrained from mal-operating during magnetising inrush by detecting the

- a) second-harmonic component of the inrush current.
- b) negative-sequence current component.
- c) zero-sequence current component.
- d) DC component only.

Answer: (a)

Question 8:

The Buchholz relay used in transformer protection is capable of detecting

- a) external short circuits.
- b) overhead-line faults.
- c) busbar faults.

d) incipient internal faults that slowly generate gas.

Answer: (d)

Question 9:

A three-phase induction motor is most commonly protected against single-phasing (loss of one supply phase) by means of a

- a) negative-sequence current relay.
- b) zero-sequence current relay.
- c) Buchholz relay.
- d) directional overcurrent relay.

Answer: (a)

Question 10:

In the differential protection of a star/delta power transformer, the current transformers are connected such that the CTs on the star-connected winding are connected in

- a) star and the CTs on the delta winding in star.
- b) delta and the CTs on the delta winding in delta.
- c) delta, while the CTs on the delta winding are connected in star.
- d) star, while the CTs on the delta winding are connected in star.

Answer: (c)

Solution:

To compensate for the 30° phase shift introduced by the power transformer, the CT connections are made opposite to the winding connection: the CTs on the star winding are connected in delta and the CTs on the delta winding are connected in star.

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